

DATA CLEANING AND TITLE OPTIMIZATION REPORT

1. INTRODUCTION

This report presents the data cleaning and title optimization process performed on a product dataset containing information such as product titles, pricing, ratings, and categories.

The primary objective of this task was to improve data quality, ensure consistency, and enhance usability by creating a concise and standardized short_title feature. This transformation enables better readability, supports search optimization, and prepares the dataset for further analysis.

2. DATASET OVERVIEW

The dataset consists of product-level information, including key variables such as:

- Product Title
- Description
- Bullet point

Initial inspection revealed that while the dataset was structured, it contained inconsistencies and noise that could affect analysis and usability.

3. DATA CLEANING PROCESS

3.1 Initial Data Exploration

A high-level review of the dataset was conducted using Excel. This included:

- Scanning column names for clarity and consistency
- Reviewing data types (text, numeric fields)
- Identifying missing values and duplicate records
- Checking for inconsistent formatting in text fields

3.2 Handling Missing Values

Columns were examined for missing or blank entries.

- Rows with missing critical information (e.g., product titles) were removed or excluded from processing
- Non-critical missing values were retained where appropriate to avoid unnecessary data loss

3.3 Removing Duplicate Entries

Duplicate records were identified and removed using Excel's "Remove Duplicates" feature. This ensured that each product entry was unique and prevented distortion in analysis.

3.4 Standardizing Data Formats

To improve consistency:

- Column names were standardized using lowercase and underscore formatting (e.g., product_title, price)
- Text fields were cleaned using trimming techniques to remove extra spaces
- Formatting inconsistencies in categorical fields were corrected

3.5 Verifying Data Accuracy

Key variables were reviewed for anomalies, including:

- Invalid or negative price values
- Inconsistent or unrealistic ratings

This step ensured that the dataset was reliable and analysis-ready.

4. SHORT TITLE CREATION

4.1 Objective

The goal of the short_title feature was to generate concise, readable, and meaningful product titles while preserving essential product information.

4.2 Methodology

The following approach was used to create optimized short titles:

- Identification of key components within product titles (brand, product type, attributes, quantity)
- Removal of redundant or non-essential phrases such as:
 - “for”, “includes”, “set of”
- Elimination of unnecessary symbols and separators (e.g., “|”, commas)
- Retention of core product identity
- Limiting title length to approximately 30–50 characters

4.3 Examples

BEFORE (Original Title)

"Small Soldiers" Official Strategy Guide (Official Strategy Guides)

AFTER (Short Title)

"Small Soldiers" Official Strategy Guide

BEFORE (Original Title)

AFTER (Short Title)

#followme 601098-GRY-XL Shorts Set for Women Grey/Black

#followme 601098-GRY-XL Shorts Set for Women Grey/Black

(BOZABOZA) 15" (Compatible with Mac Book Pro) Screen Privacy Screen Filter [Anti-Blue Ray] [Anti-Glare] [Anti-Scratch] [Matte or Gloss] Privacy Protector for Laptop, Notebook, LCD, Monitor, Display

BOZABOZA 15" MacBook Pro Privacy Screen Filter

(Renewed) CP Plus Outdoor Wireless IP Camera (2 MP)

(Renewed) CP Plus Outdoor Wireless IP Camera 2 MP

(Renewed) Orient Electric Zippy 225mm 2-in-1 Wall Mount and Table Top Fan (White/Blue)

(Renewed) Orient Electric Zippy 225mm 2-in-1 Fan

(Reprint) 1963 Yearbook: Aquinas Dominican High School, Chicago, Illinois

(Reprint) 1963 Yearbook Aquinas Dominican High School

(Reprint) 1968 Yearbook: Nixon High School, Laredo, Texas

(Reprint) 1968 Yearbook Nixon High School

(Reprint) 1972 Yearbook: Altus High School, Altus, Oklahoma

(Reprint) 1972 Yearbook Altus High School

(Reprint) 1982 Yearbook: Garinger High School, Charlotte, North Carolina

(Reprint) 1982 Yearbook Garinger High School

[3 Pack] iPhone SE 5S 5 5C Glass Screen Protector, InaRock 0.26mm 9H Tempered Glass Screen Protector for iPhone SE 5S 5 5C Most Durable [Easy-Install Wings]

3 Pack iPhone SE 5S 5 5C Glass Screen Protector

4.4 Outcome

The short_title feature significantly improves:

- Readability
- Consistency
- Suitability for display and search

5. CLEAN DATASET OVERVIEW

Following the cleaning and transformation process, the dataset demonstrates:

- Improved structural consistency
- Removal of duplicate records
- Standardized column naming conventions
- Cleaner and more usable text fields
- Addition of a value-enhancing feature (short_title)

These improvements make the dataset more suitable for analysis, reporting, and real-world applications.

6. LIMITATIONS

Despite the improvements, a few limitations remain:

- Some short titles may still require manual refinement for optimal clarity
- The dataset does not include additional contextual variables (e.g., product descriptions, brand hierarchy)
- Time constraints limited deeper validation and enrichment

7. CONCLUSION

The data cleaning and title optimization process successfully transformed the dataset into a more structured, consistent, and analysis-ready format.

The introduction of the short_title feature enhances usability and demonstrates practical feature engineering, which is essential in real-world data workflows.

Overall, this task highlights the importance of data preparation in ensuring accurate insights and effective communication.